VISVESVARAYA TECHNOLOGICAL UNIVERSITY
BELAGAVI-590018

“INTERNSHIP”
(18CSI85)

REPORT ON

“Titanic Survival Prediction Using Machine Learning”

Submitted in partial fulfillment for the requirements for the Award of Degree of

BACHELOR OF ENGINEERING
IN
INFORMATION SCIENCE AND ENGINEERING
BY

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1EP20IS070

UNDER THE GUIDANCE OF
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EAST POINT COLLEGE OF ENGINEERING & TECHNOLOGY
Department of Information Science and Engineering
Approved by AICTE New Delhi| Affiliated to VTU, Belagavi
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2023-24

EAST POINT COLLEGE OF ENGINEERING & TECHNOLOGY

(Affiliated to Visvesvaraya Technological University, Belgavi)

Bangalore-560049

DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING

CERTIFICATE

This is to certify that the **Internship** entitled **"TITANIC SURVIVAL PREDICTION USING MACHINE LEARNING"** is a bonafied work carried out by **Ms. POORNIMA B, bearing USN 1EP20IS070** in partial fulfillment for the award of **Bachelor of Engineering in Information Science and Engineering** under **Visvesvaraya Technological University, Belagavi** during the year **2023-24**. It is certified that all the corrections/suggestions indicated in the Internal Assessment have been incorporated in the report and submitted in the department library. This Internship report has been approved as it satisfies the academic requirements in respect of Internship (18CSI85) prescribed for the award of the said degree.

GUIDE	HOD	PRINCIPAL
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Asst. Professor	Dept. of ISE,	EPCET
Dept. of ISE ,EPCET	EPCET	

Examiners

Name of the Examiners **Signature with date**

1.

2.

ACKNOWLEDGEMENT

Any achievement, be it scholastic or otherwise does not depend solely on the individual efforts but on the guidance, encouragement and cooperation of intellectuals, elders and friends. I would like to take this opportunity to thank them all.

First and foremost I would like to express our sincere regards and thanks to **Mr. Pramod Gowda** and **Mr. Rajiv Gowda,** CEOs, East Point Group of Institutions, Bangalore, for providing necessary infrastructure and creating good environment.

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Name: POORNIMA B

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ABSTRACT

The sinking of RMS Titanic is presumably one of the most infamous and disastrous shipwrecks in history. During her maiden voyage and early morning hours of April 15, 1912, the Titanic regrettably sank after colliding with an iceberg, killing an approximate of 1502 passengers and crew out of 2224 making it one of many of the deadliest commercial maritime in history till date. The entire international community was deeply shocked and saddened after hearing the news of this sensational disaster which resulted in improved ship safety legislation. Her architect, Thomas Andrews died in the disaster. An eye-opening observation that came forth from the sinking of Titanic is the fact that some individuals had a better chance at surviving than the others. Kids and women had been given foremost priority. Social classes were heavily stratified in the early twentieth century, this was especially implemented on the Titanic Firstly,

the aim is use and apply exploratory data analytics (EDA) to uncover previously unknown or hidden facts in the data set available. Then the task is to later anoint various machine learning models to conclude the study of which types of individuals are more likely to live. The outcomes of application of the different machine learning models were then set side by side and analyzed based upon precision.

TABLE OF CONTENTS

LIST OF FIGURES

CERTIFICATE

Cranes Varsity

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"Where Technology Meets Excellence"

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Certificate

AICTE NO - 61216204108051630495236

Ref. No.: A/IN/263/23-24

This is to certify that Mr. / Ms. Poornima B – 1EP20IS070

of East Point College of Engineering and Technology

has successfully completed Internship

on Data Science with AIML

Conducted from 21st Aug 2023 to 22nd Sep 2023 at Cranes Varsity

Rekha.C.H

Training Manager

Authorised Signatory

CHAPTER 1

ABOUT THE COMPANY

Cranes Varsity is a pioneer Technical Training institute turned EdTech Platform offering Technology educational services for over 25 years. A division of Cranes Software International Ltd, Cranes Varsity was established with an ambitious vision of bridging the gap between the technology academia and the industry. The team continuously strives to be an organization that brings together technology and education, empowering aspiring professionals to seek assured placements and a lucrative career path. Cranes Varsity offers training to Graduates – under the Finishing School Model, Industry connects University programs, Upskilling programs for Working Professionals, and Customized training to Corporate & Defence sectors. Cranes Varsity's high- impact hands-on technology training catapults engineering students, graduates, and working professionals to be quickly employable in Niche high-end engineering Fields.

The in-house placement team further ensures that these students get placed in leading corporate firms – with whom Cranes Varsity has decades-old relationships. They stand by principle – We Assist Until We Place. Being a trusted recruitment & training partner with Corporate, they engage with them for the "Hire, Train & Deploy" Model.

The competitive advantage is tha Cranes Varsity offers an array of high-end technology training in Embedded & Automotive Systems, C, C++, MATLAB, RTOS, Linux, LDD, BSP, Embedded Testing, IoT Architecture, Protocols – Edge node Computing, Gateway & Security with industrial IoT, DSP & MATLAB, VLSI design, Java technologies, Cloud Computing, Azure, Python, Data Science & Analytics,Tableau, Artificial Intelligence with Machine Learning, Deep Learning, NLP, Business Intelligence and more The Learning Approach Model is EEE – Educate, Evolve, Employment through Pedagogical practices that integrate Learning Management Systems (LMSS). They continuously aim for participants satisfaction and placement commitment through focused Training by our Subject-Matter Experts and Professionals.

Core Values:

In performing its mission, Cranes Varsity to these core values:

- Embrace integrity and ethical conduct.
- Embrace diversity and respect the dignity and culture of all people.
- Nurture and treasure the environment and our natural and man-made resources.
- Facilitate the development, dissemination and application of engineering knowledge.
- Promote the benefits of continuing education and of engineering education.
- Respect and document engineering history while continually embracing change.
- Promote the technical and societal contribution of engineer.

Our Vision:

Cranes Varsity is to serve diverse global communities by advancing, disseminating and applying engineering knowledge for improving the quality of life; and communicating the excitement of engineering.

Our Mission:

Cranes Varsity aims to be the essential resource for mechanical, civil engineers and other technical professionals throughout the world for solutions that benefit humankind.

CHAPTER 2

OBJECTIVE OF INTERNSHIP

Internships help us to connect the dots between theory and practical work experience. The main objective of internship is to get handful of work experience during a limited time. The main purpose is to familiarize on a particular Subject or a topic that we have learned from the classrooms.

During this course I became familiar with Machine Learning application (using MATLAB & GUI) and different Algorithms of Machine Learning, Data Pre-processing. It also gives an Idea to know the applications of Machine Learning using MATLAB and where to apply them on a real time System.

From this Internship, one can develop skills in communication, Group discussion, Teamwork and many more. Moreover, we can get an overview of Industry process in product development that helped us to develop and build mini projects and converted into a product. From this, it also provides an opportunity to see how classroom and textbook learning applies to the real world.

The Student will be able to get Good moral values on responsibility, commitment, teamwork, trustworthy during training among other student.

CHAPTER 3

INTRODUCTION TO MACHINE LEARNING

3.1 Introduction

Machine Learning is the science of getting computers to learn without being explicitly programmed. It is closely related to computational statistics, which focuses on making prediction using computer. In its application across business problems, machine learning is also referred as predictive analysis. Machine Learning is closely related to computational statistics. Machine Learning focuses on the development of computer programs that can access data and use it to learn themselves. The process of learning begins with observations or data, such as examples, direct experience, or instruction, in order to look for patterns in data and make better decisions in the future based on the examples that we provide. The primary aim is to allow the computers learn automatically without human intervention or assistance and adjust actions accordingly.

3.2 History of Machine Learning

The name machine learning was coined in 1959 by Arthur Samuel. Tom M. Mitchell provided a widely quoted, more formal definition of the algorithms studied in the machine learning field: "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T, as measured by P, improves with experience E." This follows Alan Turing's proposal in his paper "Computing Machinery and Intelligence", in which the question "Can machines think?" is replaced with the question "Can machines do what we (as thinking entities) can do?". In Turing’s proposal the characteristics that could be possessed by a thinking machine and the various implications in constructing one are exposed.

3.3 The Challenges Facing Machine Learning

While there has been much progress in machine learning, there are also challenges. For example, the mainstream machine learning technologies are black-box approaches, making us concerned about their potential risks. To tackle this challenge, we may want to make machine learning more explainable and controllable. As another example, the computational complexity of machine learning algorithms is usually very high and we may want to invent lightweight algorithms or implementations. Furthermore, in many domains such as physics, chemistry, biology, and social sciences, people usually seek elegantly simple equations.

You have to gather and prepare data, then train the algorithm. There are much more uncertainties. That is why, while in traditional website or application development an experienced team can estimate the time quite precisely, a machine learning project used for example to provide product recommendations can take much less or much more time than expected. Why? Because even the best machine learning engineers don't know how the deep learning networks will behave when analyzing different sets of data. It also means that the machine learning engineers and data scientists cannot guarantee that the training process of a model can be replicated.

3.4 Types of Machine Learning

The types of machine learning algorithms differ in their approach, the type of data they input and output, and the type of task or problem that they are intended to solve. Broadly Machine Learning can be categorized into four categories.

i. Supervised Learning

ii. Unsupervised Learning

iii. Reinforcement Learning

iv. Semi-supervised Learning

Machine learning enables analysis of massive quantities of data. While it generally delivers faster, more accurate results in order to identify profitable opportunities or dangerous risks, it may also require additional time and resources to train it properly.

1. Supervised Learning

Supervised Learning is a type of learning in which we are given a data set and we already know what are correct output should look like, having the idea that there is a relationship between the input and output. Basically, it is learning task of learning a function that maps an input to an output based on example input- output pairs. It infers a function from labeled training data consisting of a set of training examples.

2. Unsupervised Learning

Unsupervised Learning is a type of learning that allows us to approach problems with little or no idea what our problem should look like. We can derive the structure by clustering the data based on a relationship among the variables in data. With unsupervised learning there is no feedback based on prediction result. Basically, it is a type of self-organized learning that helps in finding previously unknown patterns in data set without pre-existing lab.

3. Reinforcement Learning

Reinforcement learning is a learning method that interacts with its environment by producing actions and discovers errors or rewards. Trial and error search and delayed reward are the most relevant characteristics of reinforcement learning. This method allows machines and software agents to automatically determine the ideal behavior within a specific context in order to maximize its performance. Simple reward feedback is required for the agent to learn which action is best.

4. Semi-Supervised Learning

Semi-supervised learning falls somewhere in between supervised and unsupervised learning, since they use both labeled and unlabeled data for training – typically a small amount of labeled data and a large amount of unlabeled data. The systems that use this method are able to considerably improve learning accuracy. Usually, semi-supervised learning is chosen when the acquired labeled data requires skilled and relevant resources in order to train it / learn from it. Otherwise, acquiring unlabeled data generally doesn’t require additional resources.

CHAPTER 4

INTRODUCTION TO GOOGLE COLAB

What Is Google Colab?

Google Colab, or Colaboratory, is a free, cloud-based platform that lets users write and run Python code in their browser. It's an as-a-service version of Jupyter Notebook, which is a free, open-source creation from the Jupyter Project.

- Free to use
- No setup required
- Access to GPUs and TPUs
- Easy to share notebooks
- Supports multiple programming languages
- Integrated with Google Drive
- Algorithm development

Colab supports a variety of programming languages, including Python, R, and JavaScript. We can also use Colab to import data from Google Drive or other cloud storage providers.

When finished coding, you can run your code in the cloud. Colab will automatically provision the necessary resources and run your code on Google's servers.

Once code has finished running, you can view the results in the notebook. You can also export the results to a variety of formats, including CSV, JSON, and HTML.

Google Colab is a powerful tool for machine learning, data science, and education. It's free to use, easy to learn, and provides access to powerful computing resources.

CHAPTER 5

DATA PREPROCESSING, ANALYSIS & VISUALIZATION

Machine Learning algorithms don't work so well with processing raw data. Before we can feed such data to an ML algorithm, we must preprocess it. We must apply some transformations on it. With data preprocessing, we convert raw data into a clean data set. To perform data this, there are 7 techniques -

1. Rescaling Data -

For data with attributes of varying scales, we can rescale attributes to possess the same scale. We rescale attributes into the range 0 to 1 and call it normalization.

2. Standardizing Data -

With standardizing, we can take attributes with a Gaussian distribution and different means and standard deviations and transform them into a standard Gaussian distribution with a mean of 0 and a standard deviation of 1.

3. Normalizing Data -

In this task, we rescale each observation to a length of 1 (a unit norm). For this, we use the Normalizer class.

4. Binarizing Data -

Using a binary threshold, it is possible to transform our data by marking the values above it 1 and those equal to or below it, 0. For this purpose, we use the Binarizer class.

5. One Hot Encoding –

When dealing with few and scattered numerical values, we may not need to store these. Then, we can perform One Hot Encoding. For k distinct values, we can transform the feature into a k-dimensional vector with one value of 1 and 0 as the rest values.

6. Label Encoding -

Some labels can be words or numbers. Usually, training data is labelled with words to make it readable. Label encoding converts word labels into numbers to let algorithms work on them.

7. Mean Removal-

We can remove the mean from each feature to center it on zero.

CHAPTER 6

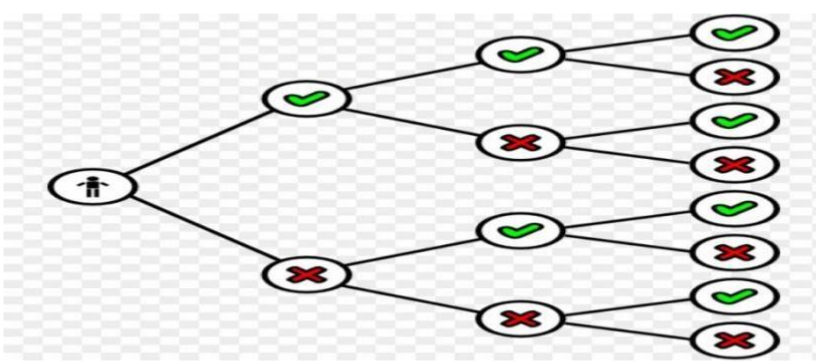
MACHINE LEARNING ALGORITHM

There are many types of Machine Learning Algorithms specific to different use cases. As we work with datasets, a machine learning algorithm works in two stages. We usually split the data around 20%-80% between testing and training stages. Under supervised learning, we split a dataset into a training data and test data in MATLAB ML. Followings are the Algorithms of Machine Learning

6.1 Decision Tree –

A decision tree falls under supervised Machine Learning Algorithms in Matlab and comes of use for both classification and regression- although mostly for classification. This model takes an instance, traverses the tree, and compares important features with a determined conditional statement. Whether it descends to the left child branch or the right depends on the result. Usually, more important features are closer to the root.

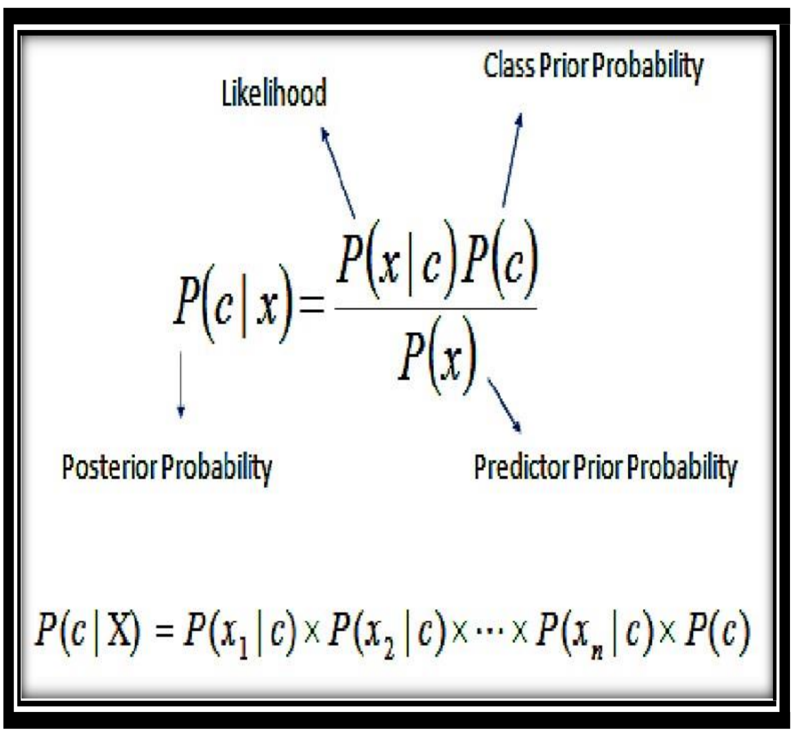
Decision Tree, a Machine Learning algorithm in Matlab can work on both categorical and continuous dependent variables. Here, we split a population into two or more homogeneous sets. Tree models where the target variable can take a discrete set of values are called classification trees; in these tree structures, leaves represent class labels and branches represent conjunctions of features that lead to those class labels. Decision trees where the target variable can take continuous values (typically real numbers) are called regression trees.



6.1. Decision Tree

6.2 Naïve Bayes Algorithm –

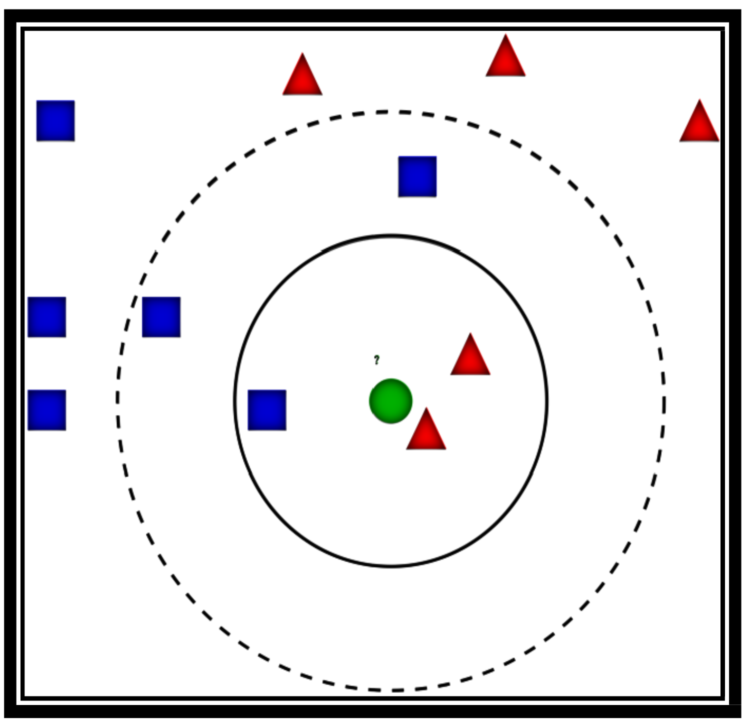
Naive Bayes is a classification method which is based on Bayes' theorem. This assumes independence between predictors. A Naive Bayes classifier will assume that a feature in a class is unrelated to any other. Consider a fruit. This is an apple if it is round, red, and 2.5 inches in diameter. A Naive Bayes classifier will say these characteristics independently contribute to the probability of the fruit being an apple. This is even if features depend on each other. For very large data sets, it is easy to build a Naive Bayesian model. Not only is this model very simple, it performs better than many highly sophisticated classification methods. Naïve Bayes classifiers are highly scalable, requiring a number of parameters linear in the number of variables (features/predictors) in a learning problem. Maximum-likelihood training can be done by evaluating a closed-form expression, which takes linear time, rather than by expensive iterative approximation as used for many other types of classifiers.



6.2 Naïve Bayes Algorithm

6.3 KNN Algorithm –

This is a Matlab Machine Learning algorithm for classification and regression- mostly for classification. This is a supervised learning algorithm that considers different centroids and uses a usually Euclidean function to compare distance. Then, it analyzes the results and classifies each point to the group to optimize it to place with all closest points to it. It classifies new cases using a majority vote of k of its neighbors. The case it assigns to a class is the one most common among its K nearest neighbors. For this, it uses a distance function. k-NN is a type of instance-based learning, or lazy learning, where the function is only approximated locally and all computation is deferred until classification. k-NN is a special case of a variable- bandwidth, kernel density "balloon" estimator with a uniform kernel.



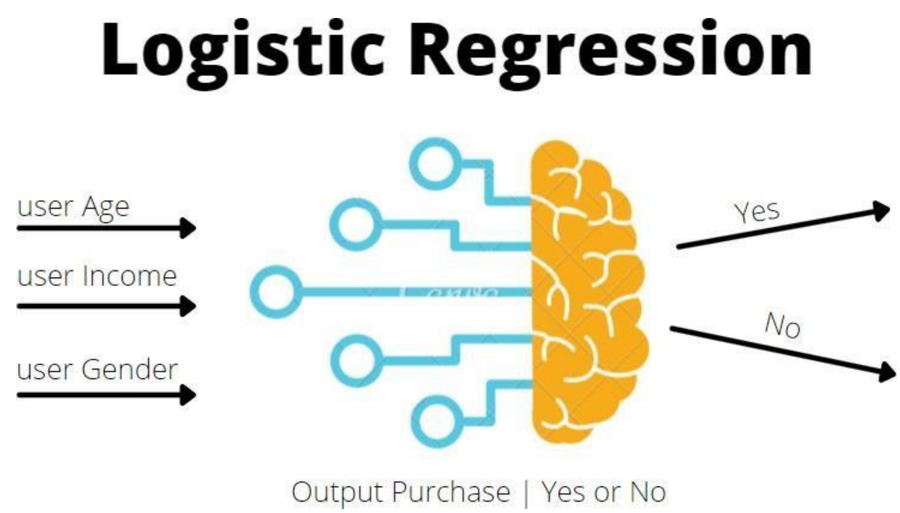
6.3. KNN Algorithm

6.4 Logistic regression:

Logistic regression is a supervised machine learning algorithm that accomplishes binary classification tasks by predicting the probability of an outcome, event, or observation. The model delivers a binary or dichotomous outcome limited to two possible outcomes: yes/no, 0/1, or true/false.

Logical regression analyzes the relationship between one or more independent variables and classifies data into discrete classes. It is extensively used in predictive modeling, where the model estimates the mathematical probability of whether an instance belongs to a specific category or not.

For example, 0 – represents a negative class; 1 – represents a positive class. Logistic regression is commonly used in binary classification problems where the outcome variable reveals either of the two categories (0 and 1).



6.4. Logistic regression

CHAPTER 7

PROS AND CONS OF MACHINE LEARNING

7.1 Advantages of Machine Learning

Every coin has two faces, each face has its own property and features. It's time to uncover the faces of ML. A very powerful tool that holds the potential to revolutionize the way things work.

1. Easily identifies trends and patterns -

Machine Learning can review large volumes of data and discover specific trends and patterns that would not be apparent to humans. For instance, for an e-commerce website like Amazon, it serves to understand the browsing behaviors and purchase histories of its users to help cater to the right products, deals, and reminders relevant to them. It uses the results to reveal relevant advertisements to them.

2. No human intervention needed (automation) -

With ML, we don't need to babysit our project every step of the way. Since it means giving machines the ability to learn, it lets them make predictions and also improve the algorithms on their own. A common example of this is anti-virus software. They learn to filter new threats as they are recognized. ML is also good at recognizing spam.

3. Continuous Improvement -

As ML algorithms gain experience, they keep improving in accuracy and efficiency. This lets them make better decisions. Say we need to make a weather forecast model. As the amount of data, we have keeps growing, our algorithms learn to make more accurate predictions faster.

4. Handling multi-dimensional and multi-variety data -

Machine Learning algorithms are good at handling data that are multi-dimensional and multi-variety, and they can do this in dynamic or uncertain environments.

7.2 Disadvantages of Machine Learning

With all those advantages to its powerfulness and popularity, Machine Learning isn't perfect. The following factors serve to limit it:

1. Data Acquisition -

Machine Learning requires massive data sets to train on, and these should be inclusive/unbiased, and of good quality. There can also be times where they must wait for new data to be generated.

2. Time and Resources -

ML needs enough time to let the algorithms learn and develop enough to fulfill their purpose with a considerable amount of accuracy and relevancy. It also needs massive resources to function. This can mean additional requirements of computer power for us.

3. Interpretation of Results -

Another major challenge is the ability to accurately interpret results generated by the algorithms. We must also carefully choose the algorithms for your purpose.

4. High error-susceptibility -

Machine Learning is autonomous but highly susceptible to errors. Suppose you train an algorithm with data sets small enough to not be inclusive. You end up with biased predictions coming from a biased training set. This leads to irrelevant advertisements being displayed to customers. In the case of ML, such blunders can set off a chain of errors that can go undetected for long periods of time. And when they do get noticed, it takes quite some time to recognize the source of the issue, and even longer to correct it.

CHAPTER 8

APPLICATIONS OF MACHINE LEARNING

Applications of Machine Learning include:

- **Web Search Engine:**

One of the reasons why search engines like google, bing etc work so well is because the system has learnt how to rank pages through a complex learning algorithm.

- **Photo tagging Applications:**

Be it Facebook or any other photo tagging application, the ability to tag friends makes it even more happening. It is all possible because of a face recognition algorithm that runs behind the application.

- **Spam Detector:**

Our mail agent like Gmail or Hotmail does a lot of hard work for us in classifying the mails and moving the spam mails to spam folder. This is again achieved by a spam classifier running in the back end of mail application.

- **Database Mining for growth of automation:**

Typical applications include Web-click data for better UX, Medical records for better automation in healthcare, biological data and many more.

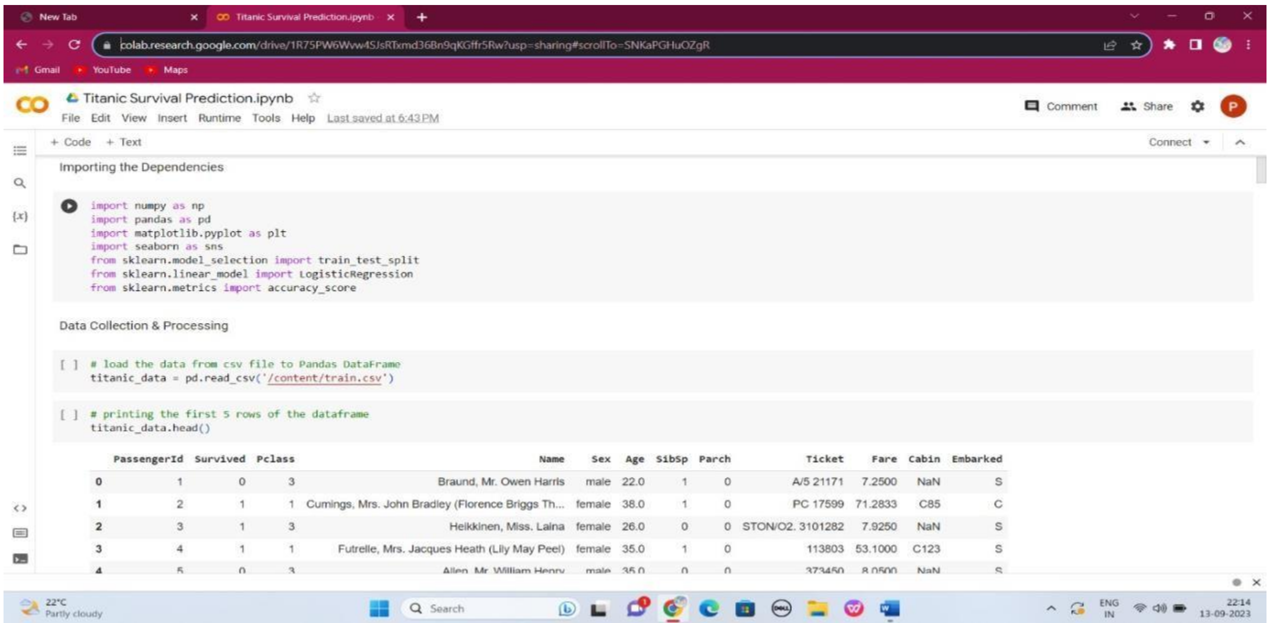
- **Applications that cannot be programmed:**

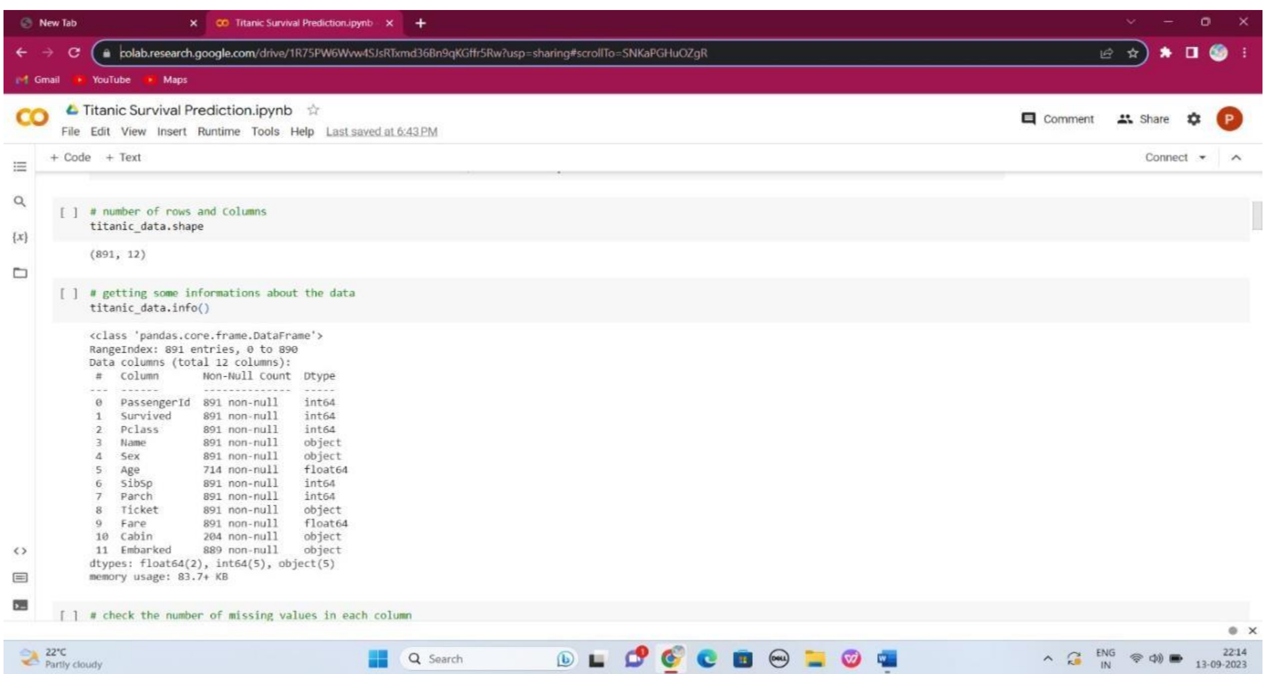
There are some tasks that cannot be programmed as the computers we use are not modelled that way. Examples include Autonomous Driving, Recognition tasks from unordered data (Face Recognition/ Handwriting Recognition), Natural language Processing, computer Vision etc.

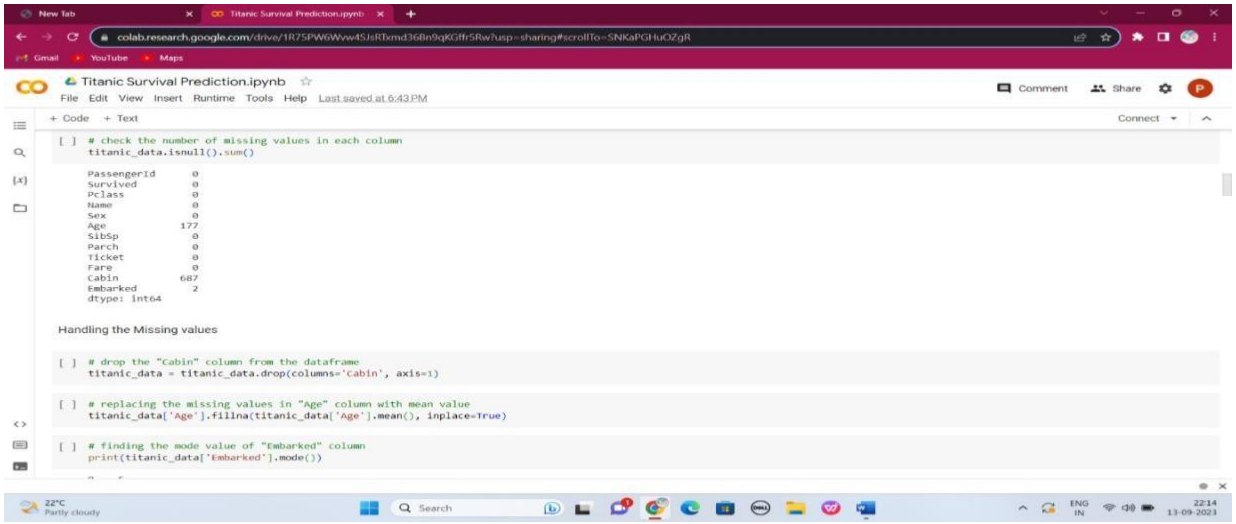
CHAPTER 9

LEARNING OUTCOME

9.1 PROJECT CODE AND OUTPUT:







Titanic Survival Prediction.ipynb

```
# check the number of missing values in each column
titanic_data.isnull().sum()
```

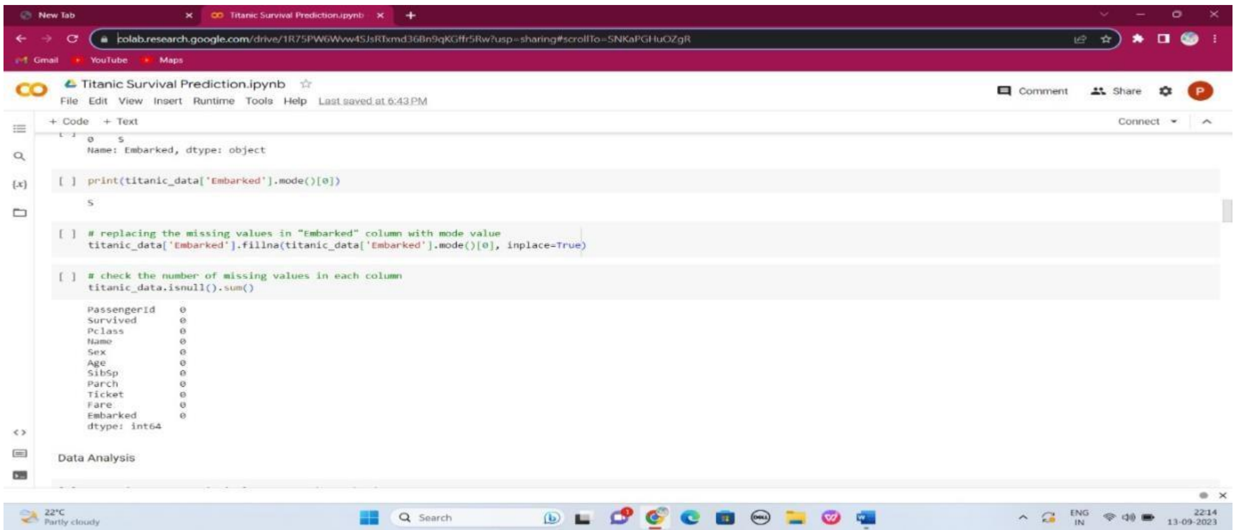
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

Handling the Missing values

```
# drop the "Cabin" column from the dataframe
titanic_data = titanic_data.drop(columns='Cabin', axis=1)
```

```
# replacing the missing values in "Age" column with mean value
titanic_data['Age'].fillna(titanic_data['Age'].mean(), inplace=True)
```

```
# finding the mode value of "Embarked" column
print(titanic_data['Embarked'].mode())
```



```
0    S
Name: Embarked, dtype: object
```

```
print(titanic_data['Embarked'].mode()[0])
```

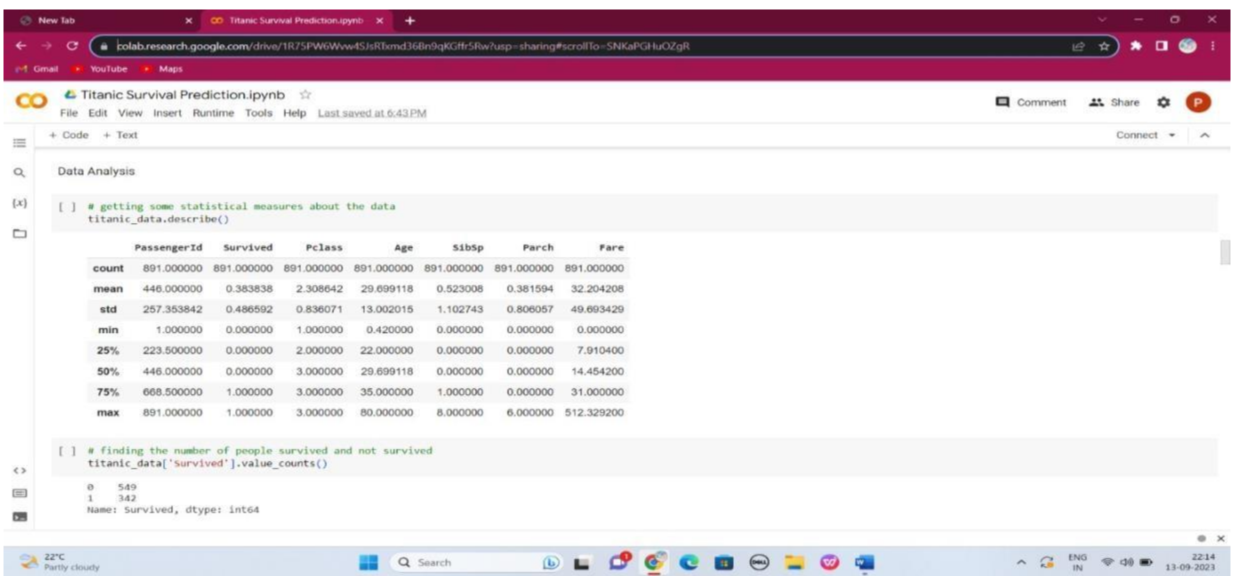
```
S
```

```
# replacing the missing values in "Embarked" column with mode value
titanic_data['Embarked'].fillna(titanic_data['Embarked'].mode()[0], inplace=True)
```

```
# check the number of missing values in each column
titanic_data.isnull().sum()
```

```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
```

Data Analysis



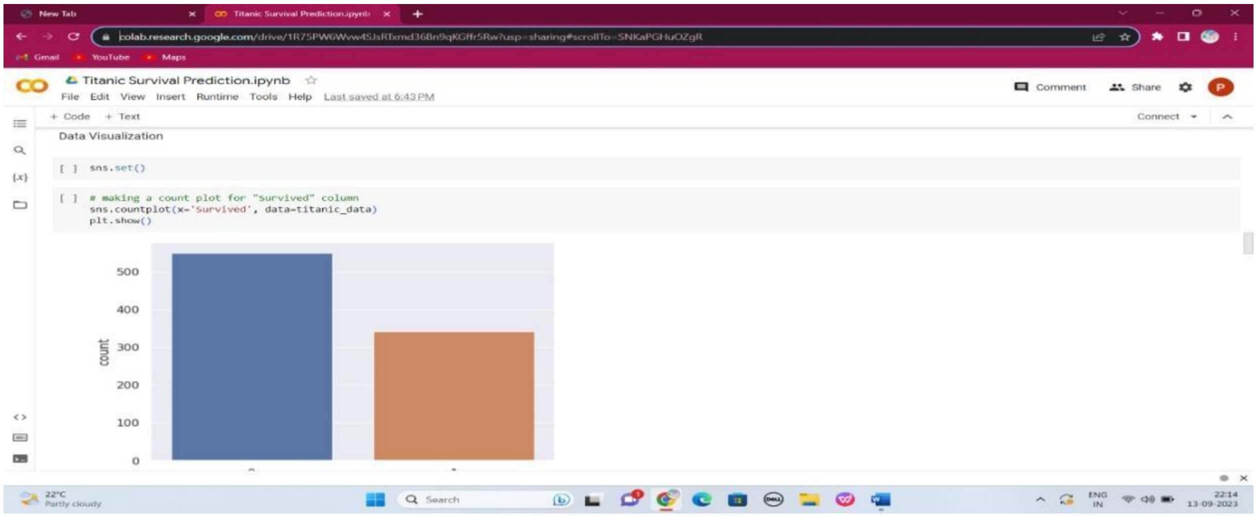
Data Analysis

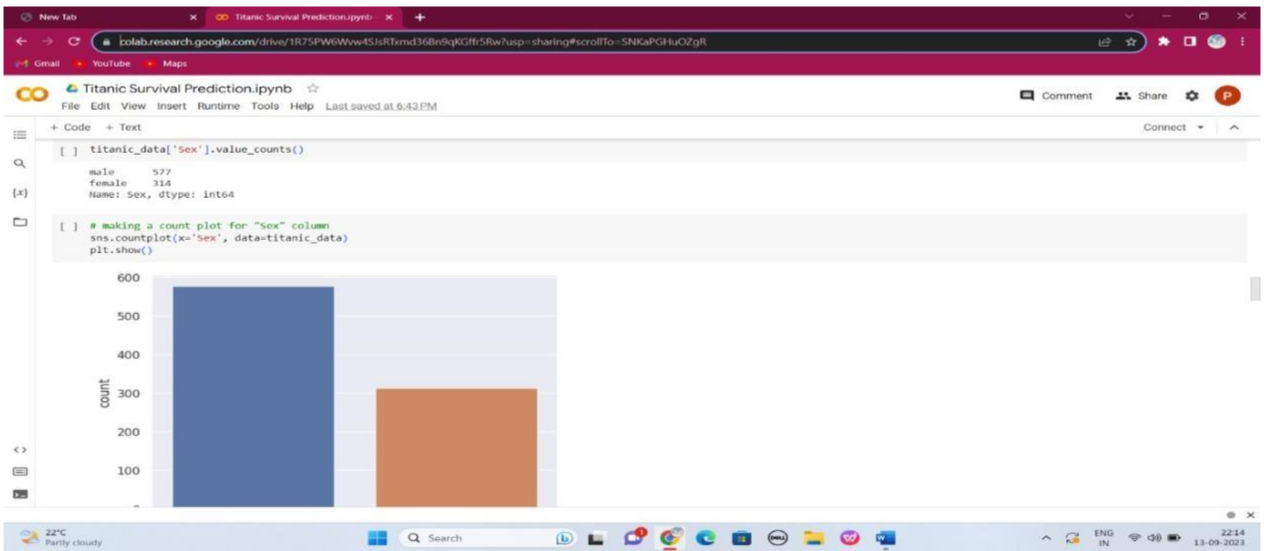
```
# getting some statistical measures about the data
titanic_data.describe()
```

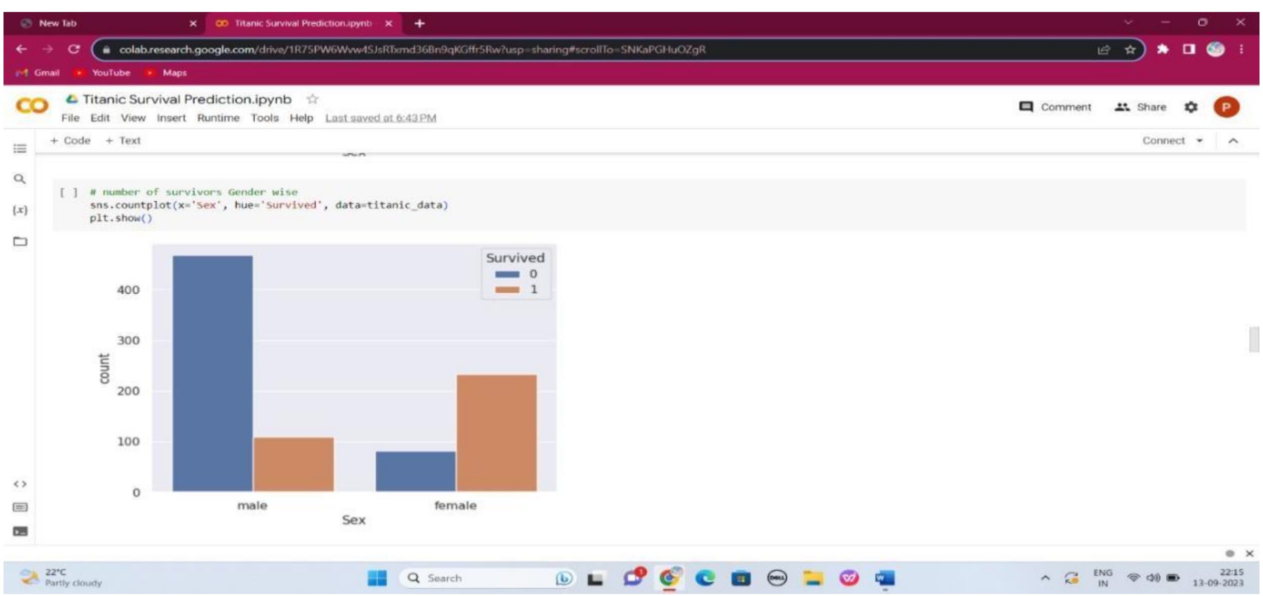
	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	891.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	13.002015	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	22.000000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	29.699118	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	35.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

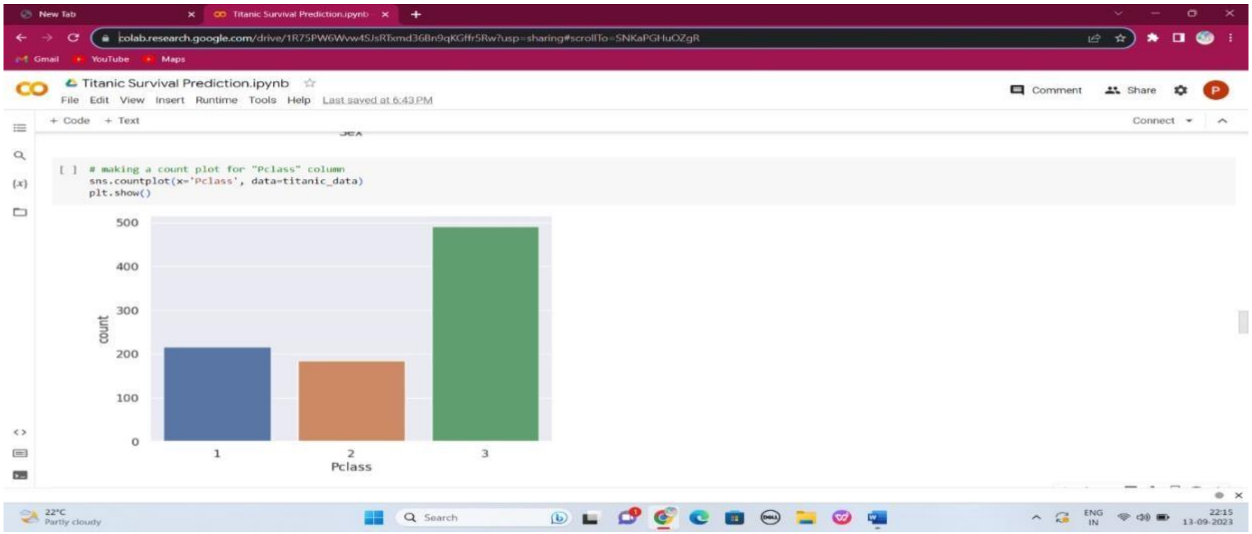
```
# finding the number of people survived and not survived
titanic_data['Survived'].value_counts()
```

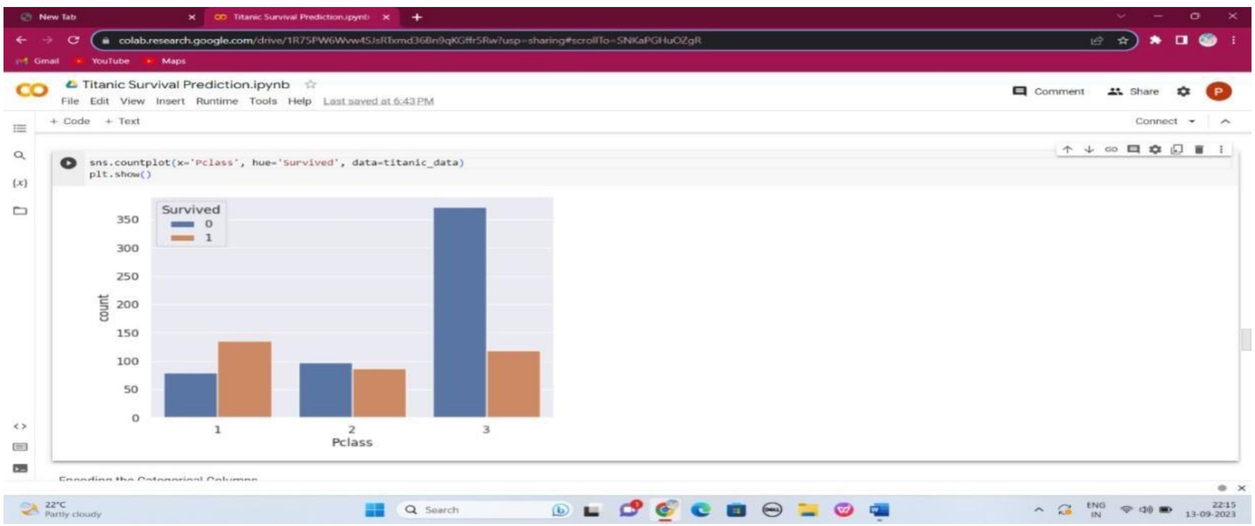
```
0    549
1    342
Name: Survived, dtype: int64
```

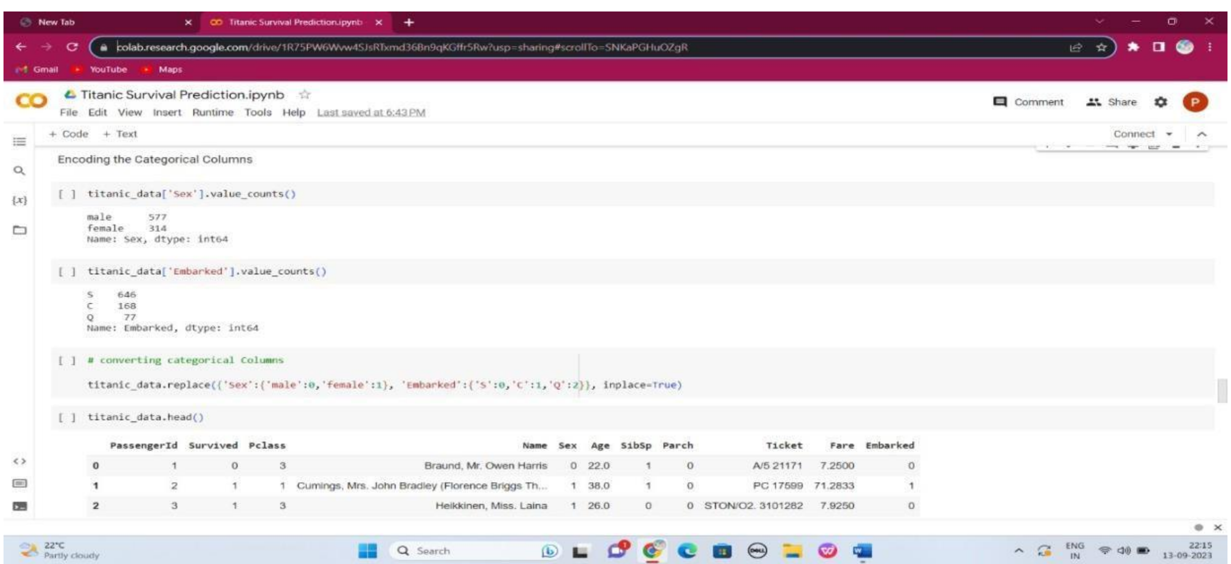

Titanic Survival Prediction.ipynb
Data Visualization
sns.set()
# making a count plot for "Survived" column
sns.countplot(x='Survived', data=titanic_data)
plt.show()
count
500
400
300
200
100
0

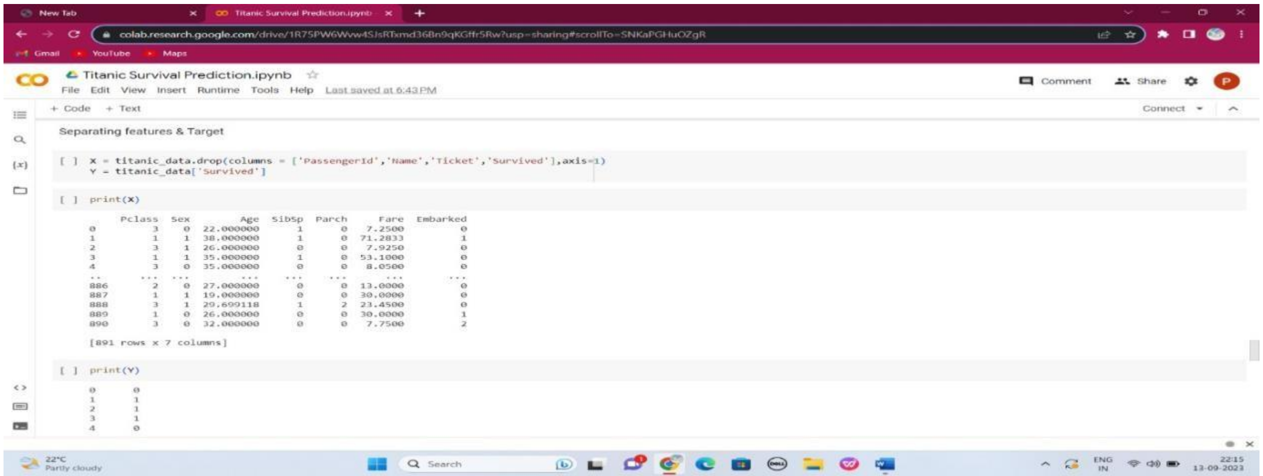

Titanic Survival Prediction.ipynb
titanic_data['Sex'].value_counts()
male 577
female 314
Name: Sex, dtype: int64
# making a count plot for "Sex" column
sns.countplot(x='Sex', data=titanic_data)
plt.show()
count
600
500
400
300
200
100

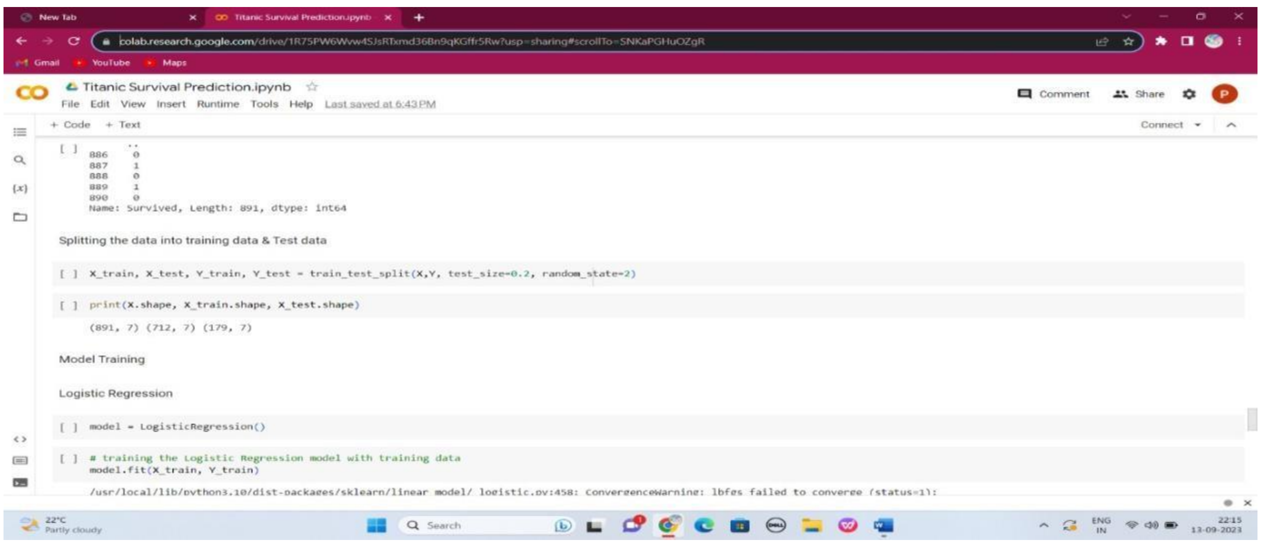

Titanic Survival Prediction.ipynb
# number of survivors Gender wise
sns.countplot(x='Sex', hue='Survived', data=titanic_data)
plt.show()
Survived
0
1
count
400
300
200
100
0
male
female
Sex

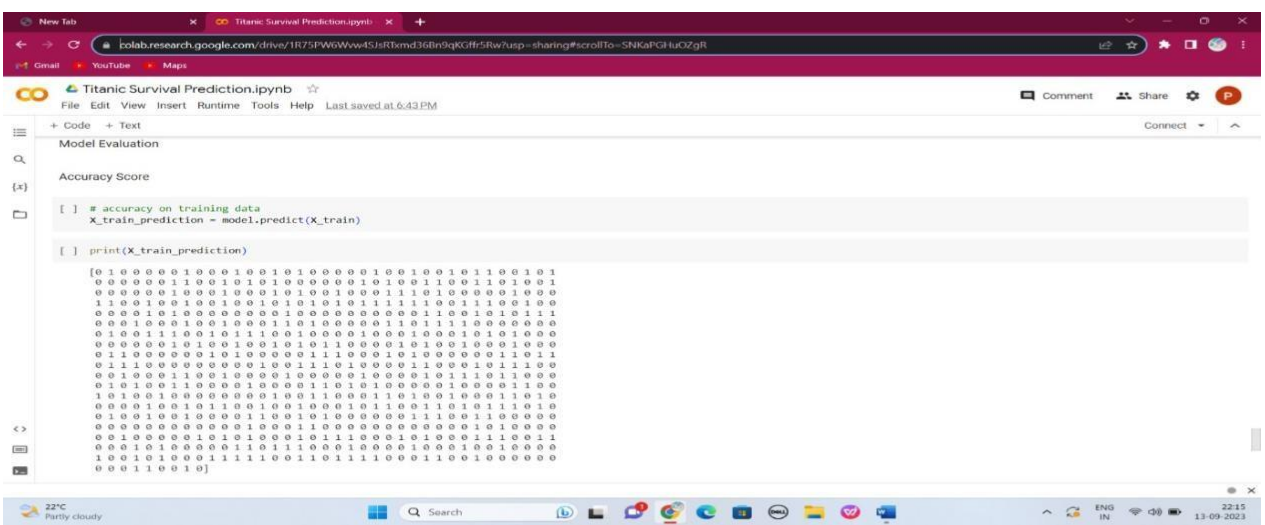

Titanic Survival Prediction.ipynb
# making a count plot for "Pclass" column
sns.countplot(x='Pclass', data=titanic_data)
plt.show()
count
Pclass

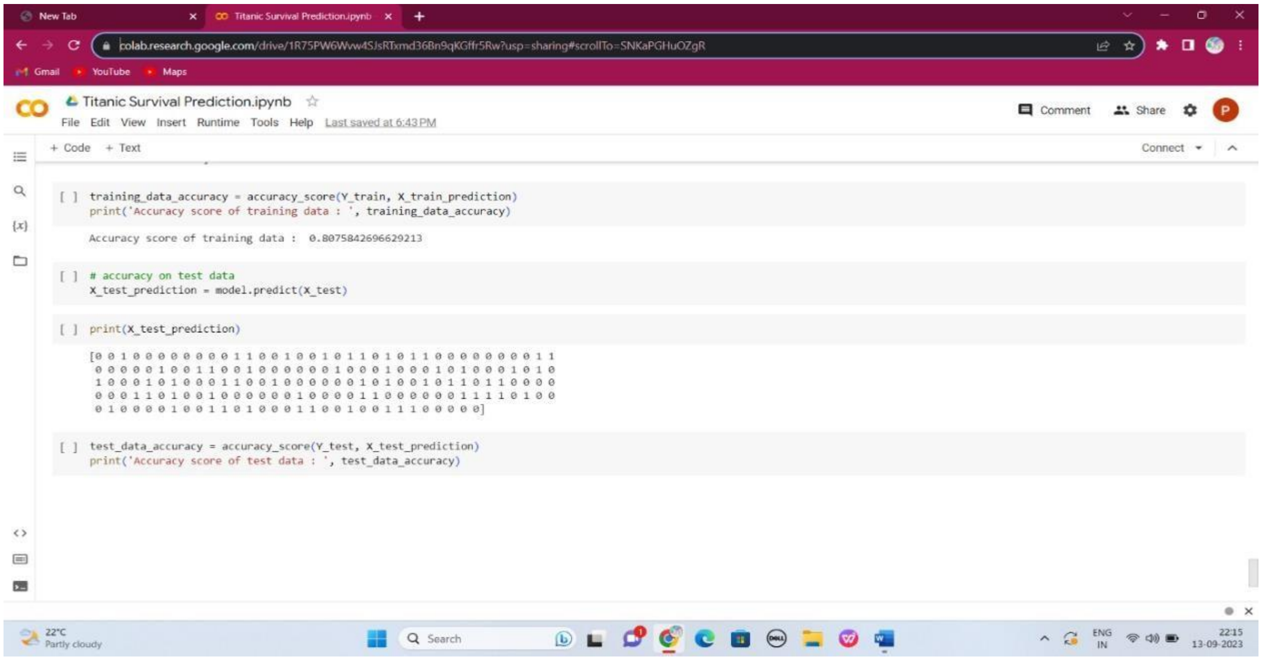

Titanic Survival Prediction.ipynb
sns.countplot(x='Pclass', hue='Survived', data=titanic_data)
plt.show()
Survived
count
Pclass


Titanic Survival Prediction.ipynb
Encoding the Categorical Columns
titanic_data['Sex'].value_counts()
male 577
female 314
Name: Sex, dtype: int64
titanic_data['Embarked'].value_counts()
S 646
C 168
Q 77
Name: Embarked, dtype: int64
# converting categorical Columns
titanic_data.replace({'Sex':{'male':0,'female':1}, 'Embarked':{'S':0,'C':1,'Q':2}}, inplace=True)
titanic_data.head()
PassengerId Survived Pclass Name Sex Age SibSp Parch Ticket Fare Embarked
0 1 0 3 Braund, Mr. Owen Harris 0 22.0 1 0 A/5 21171 7.2500 0
1 2 1 1 Cumings, Mrs. John Bradley (Florence Briggs Th... 1 38.0 1 0 PC 17599 71.2833 1
2 3 1 3 Heikkinen, Miss. Laina 1 26.0 0 0 STON/O2. 3101282 7.9250 0

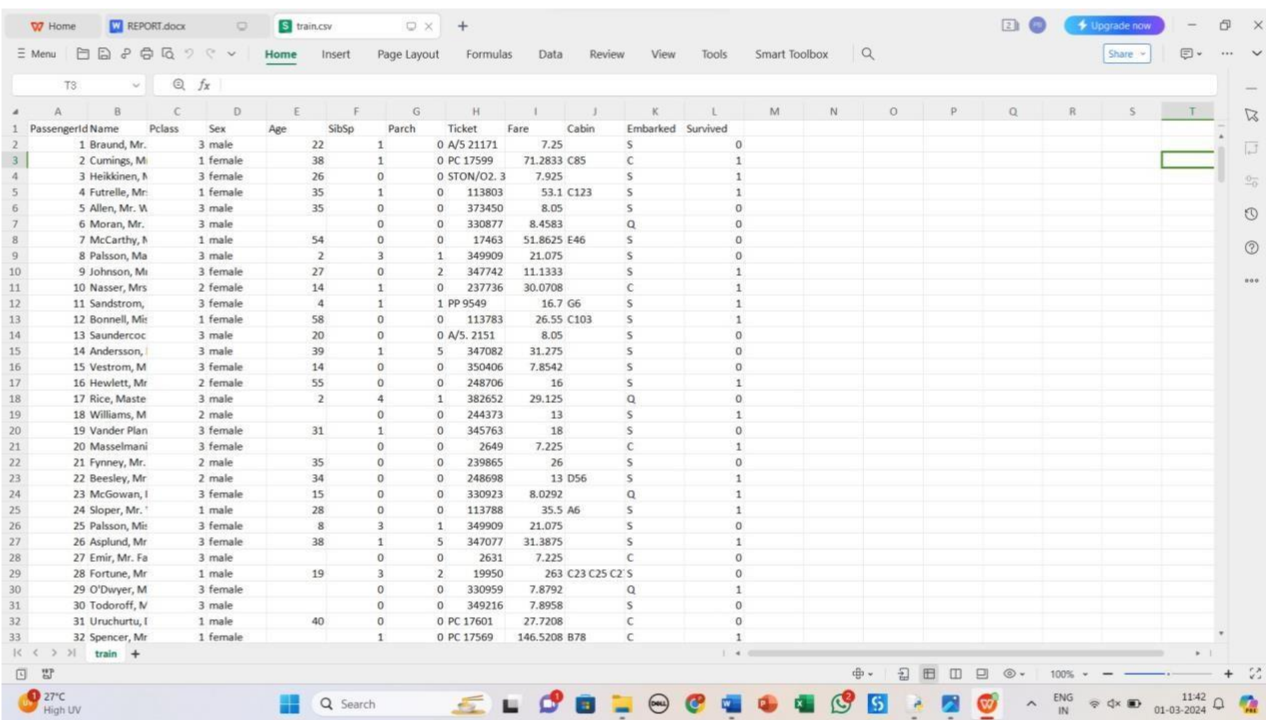

Titanic Survival Prediction.ipynb
Separating features & Target
X = titanic_data.drop(columns = ['PassengerId','Name','Ticket','Survived'],axis=1)
Y = titanic_data['Survived']
print(X)
Pclass Sex Age SibSp Parch Fare Embarked
0 3 0 22.000000 1 0 7.2500 0
1 1 1 38.000000 1 0 71.2833 1
2 3 1 26.000000 0 0 7.9250 0
3 1 1 35.000000 1 0 53.1000 0
4 3 0 35.000000 0 0 8.0500 0
886 2 0 27.000000 0 0 13.0000 0
887 1 1 19.000000 0 0 30.0000 0
888 3 1 29.699118 1 2 23.4500 0
889 1 0 26.000000 0 0 30.0000 1
890 3 0 32.000000 0 0 7.7500 2
[891 rows x 7 columns]
print(Y)


Titanic Survival Prediction.ipynb
Name: Survived, Length: 891, dtype: int64
Splitting the data into training data & Test data
X_train, X_test, Y_train, Y_test = train_test_split(X,Y, test_size=0.2, random_state=2)
print(X.shape, X_train.shape, X_test.shape)
(891, 7) (712, 7) (179, 7)
Model Training
Logistic Regression
model = LogisticRegression()
# training the Logistic Regression model with training data
model.fit(X_train, Y_train)

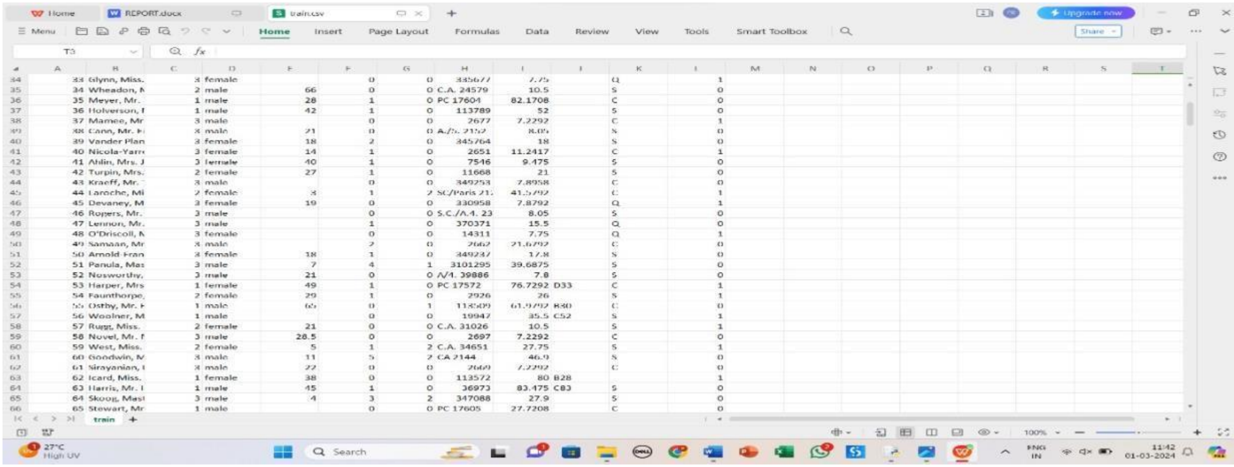

Titanic Survival Prediction.ipynb
Model Evaluation
Accuracy Score
# accuracy on training data
X_train_prediction = model.predict(X_train)
print(X_train_prediction)



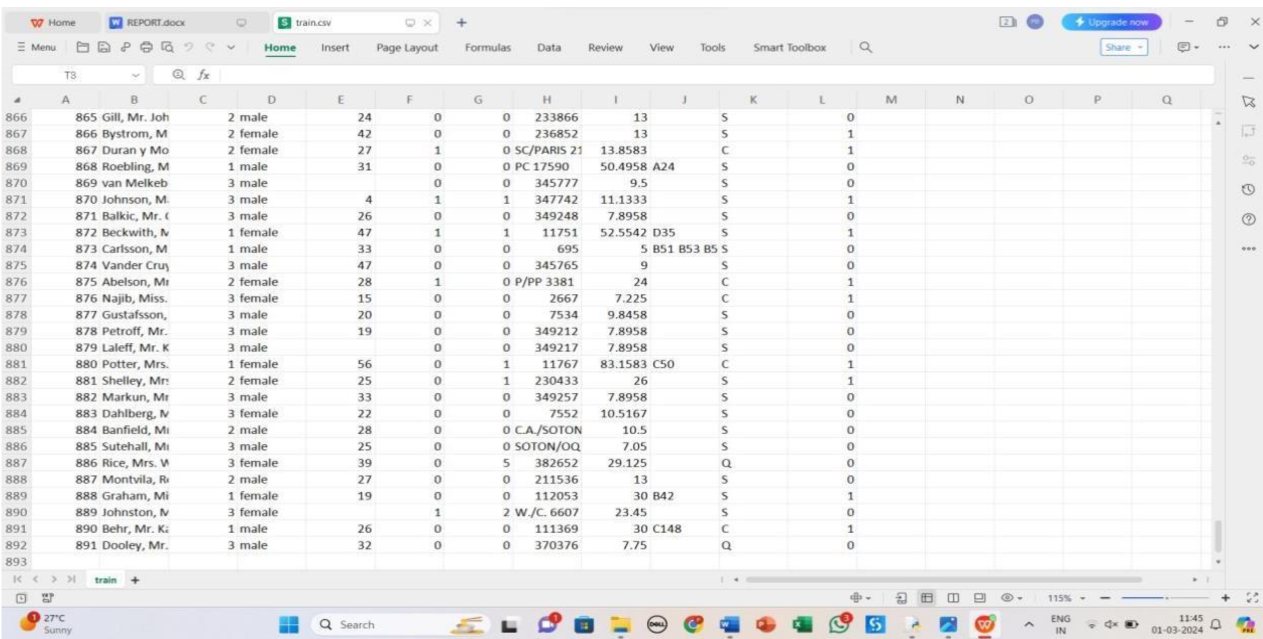
9.2 SOME RANDOM SAMPLE DATA THAT IS BEEN TRAINED



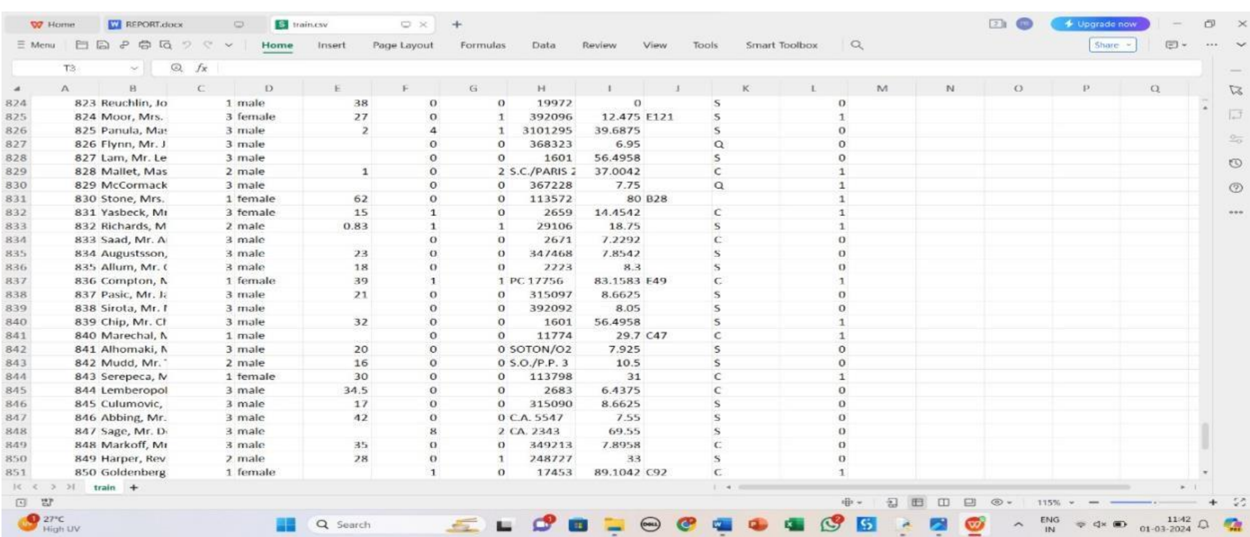
PassengerId	Name	Pclass	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Survived
1	Braund, Mr.	3	male	22	1	0	A/5 21171	7.25		S	0
2	Cumings, Mi	1	female	38	1	0	PC 17599	71.2833	C85	C	1
3	Heikkinen, N	3	female	26	0	0	STON/O2. 3	7.925		S	1
4	Futrelle, Mr	1	female	35	1	0	113803	53.1	C123	S	1
5	Allen, Mr. W	3	male	35	0	0	373450	8.05		S	0
6	Moran, Mr.	3	male		0	0	330877	8.4583		Q	0
7	McCarthy, N	1	male	54	0	0	17463	51.8625	E46	S	0
8	Palsson, Ma	3	male	2	3	1	349909	21.075		S	0
9	Johnson, Mi	3	female	27	0	2	347742	11.1333		S	1
10	Nasser, Mrs	2	female	14	1	0	237736	30.0708		C	1
11	Sandstrom,	3	female	4	1	1	PP 9549	16.7	G6	S	1
12	Bonnell, Mis	1	female	58	0	0	113783	26.55	C103	S	1
13	Saundercoc	3	male	20	0	0	A/5. 2151	8.05		S	0
14	Andersson,	3	male	39	1	5	347082	31.275		S	0
15	Vestrom, M	3	female	14	0	0	350406	7.8542		S	0
16	Hewlett, Mr	2	female	55	0	0	248706	16		S	1
17	Rice, Maste	3	male	2	4	1	382652	29.125		Q	0
18	Williams, M	2	male		0	0	244373	13		S	1
19	Vander Plan	3	female	31	1	0	345763	18		S	0
20	Masselmani	3	female		0	0	2649	7.225		C	1
21	Fynney, Mr.	2	male	35	0	0	239865	26		S	0
22	Beesley, Mr	2	male	34	0	0	248698	13	D56	S	1
23	McGowan, I	3	female	15	0	0	330923	8.0292		Q	1
24	Sloper, Mr.	1	male	28	0	0	113788	35.5	A6	S	1
25	Palsson, Mis	3	female	8	3	1	349909	21.075		S	0
26	Asplund, Mr	3	female	38	1	5	347077	31.3875		S	1
27	Emir, Mr. Fa	3	male		0	0	2631	7.225		C	0
28	Fortune, Mr	1	male	19	3	2	19950	263	C23 C25 C2	S	0
29	O'Dwyer, M	3	female		0	0	330959	7.8792		Q	1
30	Todoroff, M	3	male		0	0	349216	7.8958		S	0
31	Uruchurtu, I	1	male	40	0	0	PC 17601	27.7208		C	0
32	Spencer, Mr	1	female		1	0	PC 17569	146.5208	B78	C	1



	A	B	C	D	E	F	G	H	I	J	K	L
34	33	Glynn, Miss.	3	female		0	0	335677	7.75		Q	1
35	34	Wheadon, M	2	male	66	0	0	C.A. 24579	10.5		S	0
36	35	Meyer, Mr.	1	male	28	1	0	PC 17604	82.1708		C	0
37	36	Holverson, I	1	male	42	1	0	113789	52		S	0
38	37	Mamee, Mr	3	male		0	0	2677	7.2292		C	1
39	38	Cann, Mr. E	3	male	21	0	0	A./5. 2152	8.05		S	0
40	39	Vander Plan	3	female	18	2	0	345764	18		S	0
41	40	Nicola-Yarr	3	female	14	1	0	2651	11.2417		C	1
42	41	Ahlin, Mrs. J	3	female	40	1	0	7546	9.475		S	0
43	42	Turpin, Mrs.	2	female	27	1	0	11668	21		S	0
44	43	Kraeff, Mr.	3	male		0	0	349253	7.8958		C	0
45	44	Laroche, Mi	2	female	3	1	2	SC/Paris 21	41.5792		C	1
46	45	Devaney, M	3	female	19	0	0	330958	7.8792		Q	1
47	46	Rogers, Mr.	3	male		0	0	S.C./A.4. 23	8.05		S	0
48	47	Lennon, Mr.	3	male		1	0	370371	15.5		Q	0
49	48	O'Driscoll, N	3	female		0	0	14311	7.75		Q	1
50	49	Samaan, Mr	3	male		2	0	2662	21.6792		C	0
51	50	Arnold-Fran	3	female	18	1	0	349237	17.8		S	0
52	51	Panula, Mas	3	male	7	4	1	3101295	39.6875		S	0
53	52	Nosworthy,	3	male	21	0	0	A/4. 39886	7.8		S	0
54	53	Harper, Mrs	1	female	49	1	0	PC 17572	76.7292	D33	C	1
55	54	Faunthorpe,	2	female	29	1	0	2926	26		S	1
56	55	Ostby, Mr. E	1	male	65	0	1	113509	61.9792	B30	C	0
57	56	Woolner, M	1	male		0	0	19947	35.5	C52	S	1
58	57	Rugg, Miss.	2	female	21	0	0	C.A. 31026	10.5		S	1
59	58	Novel, Mr. I	3	male	28.5	0	0	2697	7.2292		C	0
60	59	West, Miss.	2	female	5	1	2	C.A. 34651	27.75		S	1
61	60	Goodwin, M	3	male	11	5	2	CA 2144	46.9		S	0
62	61	Sirayanian, I	3	male	22	0	0	2669	7.2292		C	0
63	62	Icard, Miss.	1	female	38	0	0	113572	80	B28		1
64	63	Harris, Mr. I	1	male	45	1	0	36973	83.475	C83	S	0
65	64	Skoog, Mast	3	male	4	3	2	347088	27.9		S	0
66	65	Stewart, Mr	1	male		0	0	PC 17605	27.7208		C	0



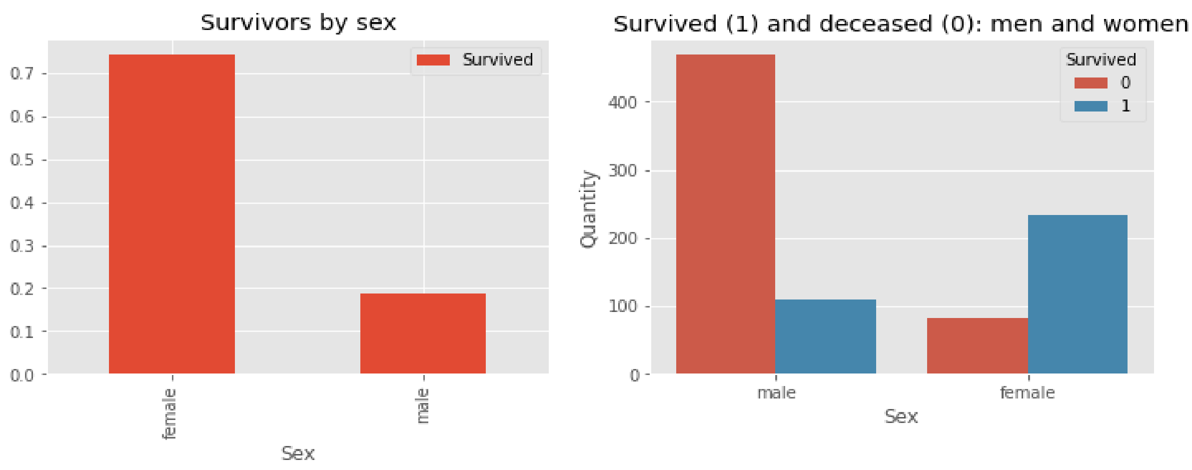
	A	B	C	D	E	F	G	H	I	J	K	L
866	865	Gill, Mr. Joh	2	male	24	0	0	233866	13		S	0
867	866	Bystrom, M	2	female	42	0	0	236852	13		S	1
868	867	Duran y Mo	2	female	27	1	0	SC/PARIS 21	13.8583		C	1
869	868	Roebling, M	1	male	31	0	0	PC 17590	50.4958	A24	S	0
870	869	van Melkeb	3	male		0	0	345777	9.5		S	0
871	870	Johnson, M.	3	male	4	1	1	347742	11.1333		S	1
872	871	Balkic, Mr. (	3	male	26	0	0	349248	7.8958		S	0
873	872	Beckwith, M	1	female	47	1	1	11751	52.5542	D35	S	1
874	873	Carlsson, M	1	male	33	0	0	695	5	B51 B53 B5	S	0
875	874	Vander Cruy	3	male	47	0	0	345765	9		S	0
876	875	Abelson, Mi	2	female	28	1	0	P/PP 3381	24		C	1
877	876	Najib, Miss.	3	female	15	0	0	2667	7.225		C	1
878	877	Gustafsson,	3	male	20	0	0	7534	9.8458		S	0
879	878	Petroff, Mr.	3	male	19	0	0	349212	7.8958		S	0
880	879	Laleff, Mr. K	3	male		0	0	349217	7.8958		S	0
881	880	Potter, Mrs.	1	female	56	0	1	11767	83.1583	C50	C	1
882	881	Shelley, Mrs	2	female	25	0	1	230433	26		S	1
883	882	Markun, Mr	3	male	33	0	0	349257	7.8958		S	0
884	883	Dahlberg, M	3	female	22	0	0	7552	10.5167		S	0
885	884	Banfield, Mr	2	male	28	0	0	C.A./SOTON	10.5		S	0
886	885	Sutehall, Mr	3	male	25	0	0	SOTON/OQ	7.05		S	0
887	886	Rice, Mrs. W	3	female	39	0	5	382652	29.125		Q	0
888	887	Montvila, R	2	male	27	0	0	211536	13		S	0
889	888	Graham, Mi	1	female	19	0	0	112053	30	B42	S	1
890	889	Johnston, M	3	female		1	2	W./C. 6607	23.45		S	0
891	890	Behr, Mr. K	1	male	26	0	0	111369	30	C148	C	1
892	891	Dooley, Mr.	3	male	32	0	0	370376	7.75		Q	0
893												



	A	B	C	D	E	F	G	H	I	J	K	L
824	823	Reuchlin, Jo	1	male	38	0	0	19972	0		S	0
825	824	Moor, Mrs.	3	female	27	0	1	392096	12.475	E121	S	1
826	825	Panula, Mas	3	male	2	4	1	3101295	39.6875		S	0
827	826	Flynn, Mr. J	3	male		0	0	368323	6.95		Q	0
828	827	Lam, Mr. Le	3	male		0	0	1601	56.4958		S	0
829	828	Mallet, Mas	2	male	1	0	2	S.C./PARIS 2	37.0042		C	1
830	829	McCormack	3	male		0	0	367228	7.75		Q	1
831	830	Stone, Mrs.	1	female	62	0	0	113572	80	B28		1
832	831	Yasbeck, Mr	3	female	15	1	0	2659	14.4542		C	1
833	832	Richards, M	2	male	0.83	1	1	29106	18.75		S	1
834	833	Saad, Mr. A	3	male		0	0	2671	7.2292		C	0
835	834	Augustsson,	3	male	23	0	0	347468	7.8542		S	0
836	835	Allum, Mr. (	3	male	18	0	0	2223	8.3		S	0
837	836	Compton, N	1	female	39	1	1	PC 17756	83.1583	E49	C	1
838	837	Pasic, Mr. Ja	3	male	21	0	0	315097	8.6625		S	0
839	838	Sirota, Mr. I	3	male		0	0	392092	8.05		S	0
840	839	Chip, Mr. Cl	3	male	32	0	0	1601	56.4958		S	1
841	840	Marechal, N	1	male		0	0	11774	29.7	C47	C	1
842	841	Alhomaki, N	3	male	20	0	0	SOTON/O2	7.925		S	0
843	842	Mudd, Mr.	2	male	16	0	0	S.O./P.P. 3	10.5		S	0
844	843	Serepeca, M	1	female	30	0	0	113798	31		C	1
845	844	Lemberopol	3	male	34.5	0	0	2683	6.4375		C	0
846	845	Culumovic,	3	male	17	0	0	315090	8.6625		S	0
847	846	Abbing, Mr.	3	male	42	0	0	C.A. 5547	7.55		S	0
848	847	Sage, Mr. D	3	male		8	2	CA. 2343	69.55		S	0
849	848	Markoff, Mr	3	male	35	0	0	349213	7.8958		C	0
850	849	Harper, Rev	2	male	28	0	1	248727	33		S	0
851	850	Goldenberg	1	female		1	0	17453	89.1042	C92	C	1

9.3 OUTPUT:

This graph shows the survivors in titanic tragedy



9.3. Output

CONCLUSION

In order to further improve the overall result, an extensive hyperparameter should be tuned on several machine learning models. It is possible to further improve it by doing ensemble learning. This research paper began with data exploration, and subsequently led to checking about absent data and learning what features are important. At the arrival of the data preprocessing portion, missing values were computed and converted features into numeric ones. Later a few more features were created. 8 different machine learning models were also simultaneously trained thereby and applied cross validation onto the chosen Random Forest model. Finally, the confusion matrix was looked into and evaluated as well as the f-score, computer model's precision and recall. When attempting data analysis, data cleaning is the first and foremost step. Exploratory data analytics allows one to recognize the dataset and the dependency linking characteristics. EDA is implemented in order to find out the connection between the features of the dataset. This is achieved with the use of different graphical techniques. Some conclusions are drawn by applying EDA and the facts are discovered.

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